


```

clear; clc; close all;

% COLLECTED LEAF DATA REPOSITORY
% Format: {'Common Name', 'Scientific Name', 'Type', 'Venation', 'Margin', 'Shape', 'Color', 'Filename'}
leaf_data = {
    'Avocado', 'Persea americana', 'Simple', 'Reticulate', 'Entire', 'Elliptic', 'Green', 'avocado leaf.png';
    'Banana', 'Musa acuminata', 'Simple', 'Parallel', 'Entire', 'Oblong', 'Green', 'Banana leaf.png';
    'Bean', 'Phaseolus vulgaris', 'Compound', 'Reticulate', 'Entire', 'Ovate', 'Green', 'bean leaf.png';
    'Bougainvillea', 'Bougainvillea glabra', 'Simple', 'Reticulate', 'Entire', 'Ovate', 'Green', 'Bougainvillea leaf.png';
    'Cassava', 'Manihot esculenta', 'Simple', 'Reticulate', 'Lobed', 'Palmate', 'Green', 'cassava leaf.png';
    'Cassia', 'Senna siamea', 'Compound', 'Reticulate', 'Entire', 'Elliptic', 'Green', 'cassia leaf.png';
    'Coffee', 'Coffea robusta', 'Simple', 'Reticulate', 'Entire', 'Elliptic', 'Dark Green', 'coffee leaf.png';
    'Eucalyptus', 'Eucalyptus globulus', 'Simple', 'Reticulate', 'Entire', 'Lanceolate', 'Grey-Green', 'eucalyptus leaf.png';
    'Guava', 'Psidium guajava', 'Simple', 'Reticulate', 'Entire', 'Elliptic', 'Green', 'guava leaf.png';
    'Hibiscus', 'Hibiscus rosa-sinensis', 'Simple', 'Reticulate', 'Serrated', 'Ovate', 'Dark Green', 'Hibiscus leaf.png';
    'Jackfruit', 'Artocarpus heterophyllus', 'Simple', 'Reticulate', 'Entire', 'Elliptic', 'Dark Green', 'jack fruit leaf.png';
    'Maize', 'Zea mays', 'Simple', 'Parallel', 'Entire', 'Linear', 'Green', 'maize leaf.png';
    'Mango', 'Mangifera indica', 'Simple', 'Reticulate', 'Entire', 'Oval', 'Green', 'mango leaf.png';
    'Neem', 'Azadirachta indica', 'Compound', 'Reticulate', 'Serrated', 'Lanceolate', 'Green', 'neem leaf.png';
    'Orange', 'Citrus sinensis', 'Simple', 'Reticulate', 'Entire', 'Ovate', 'Green', 'orange leaf.png';
    'Pawpaw', 'Carica papaya', 'Simple', 'Reticulate', 'Lobed', 'Palmate', 'Green', 'PawPaw leaf.png';
    'Rose', 'Rosa sinensis', 'Compound', 'Reticulate', 'Serrated', 'Oval', 'Green', 'rose leaf.png';
    'Sugarcane', 'Saccharum officinarum', 'Simple', 'Parallel', 'Entire', 'Linear', 'Green', 'sugarcane leaf.png';
    'Sweet Potato', 'Ipomoea batatas', 'Simple', 'Reticulate', 'Lobed', 'Cordate', 'Green', 'sweet potato leaf.png';
    'Tomato', 'Solanum lycopersicum', 'Compound', 'Reticulate', 'Lobed', 'Ovate', 'Green', 'Tomato leaf.png'
};

% BASE FOLDER FOR THE IMAGES
baseFolder = "C:\Users\Admin\OneDrive\Desktop\matlab leaf works";

% Initialize the structure array to prevent dynamic resizing overhead
leaves = struct('commonName', {}, 'scientificName', {}, 'leafType', {}, 'venation', {}, ...
    'margin', {}, 'shape', {}, 'color', {}, 'originalImage', {}, ...
    'grayImage', {}, 'binaryImage', {}, 'edgeImage', {}, 'resizedImage', {});

% IMAGE PROCESSING & PLOTTING
for i = 1:size(leaf_data, 1)
    % Assign text data attributes
    leaves(i).commonName = leaf_data{i,1};
    leaves(i).scientificName = leaf_data{i,2};
    leaves(i).leafType = leaf_data{i,3};
    leaves(i).venation = leaf_data{i,4};
    leaves(i).margin = leaf_data{i,5};
    leaves(i).shape = leaf_data{i,6};
    leaves(i).color = leaf_data{i,7};

    % Unified file path
    imgPath = fullfile(baseFolder, leaf_data{i,8});

    % Check if file exists before processing
    if ~exist(imgPath, 'file')
        warning('File not found, skipping: %s', imgPath);
        continue;
    end

    % Processing image matrix dynamically from folder
    leaves(i).originalImage = imread(imgPath);
    leaves(i).grayImage = rgb2gray(leaves(i).originalImage);
    leaves(i).binaryImage = imbinarize(leaves(i).grayImage);
    leaves(i).edgeImage = edge(leaves(i).grayImage, 'Canny');
    leaves(i).resizedImage = imresize(leaves(i).originalImage, 0.5);

    % Matrix visualization window
    fig = figure('Name', ['Results - ' leaves(i).commonName], 'Visible', 'off');

    subplot(2,3,1); imshow(leaves(i).originalImage); title('Original');
    subplot(2,3,2); imshow(leaves(i).grayImage); title('Gray Scale');
    subplot(2,3,3); imshow(leaves(i).binaryImage); title('Binary');
    subplot(2,3,4); imshow(leaves(i).edgeImage); title('Edges (Canny)');
    subplot(2,3,5); imshow(leaves(i).resizedImage); title('Resized (0.5)');
    subplot(2,3,6); histogram(leaves(i).grayImage); title('Gray Histogram');
    sgtitle(['Image Processing Sequence: ' leaves(i).commonName ' Leaf']);

    % Processed graphic report layout
    outputFileName = ['results_' lower(strrep(leaves(i).commonName, ' ', '_')) '.png'];
    saveas(fig, fullfile(baseFolder, outputFileName));
    close(fig);
end

```

Warning: File not found, skipping: C:\Users\Admin\OneDrive\Desktop\matlab leaf works\Bougainvillea

Warning: File not found, skipping: C:\Users\Admin\OneDrive\Desktop\matlab leaf works\bungainville
leaf.png
Warning: File not found, skipping: C:\Users\Admin\OneDrive\Desktop\matlab leaf works\jack fruit
leaf.png

```
% SAVE STRUCT TO FILE  
save(fullfile(baseFolder, 'matlab_leaveswork_Struct.mat'), 'leaves');  
disp('Processed all available PNG leaves successfully. matlab_leaveswork_Struct.mat .');
```

Processed all available PNG leaves successfully. matlab_leaveswork_Struct.mat .